

The status of academic research on salmon farming: a living evidence map

Results

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Review process

The review process is outlined in the flow diagram in Figure 1. Across the 5 databases searched, a total of 90,137 search results was obtained. Following deduplication, 32,292 unique candidate records remained. Title and abstract repair and enrichment supplemented 2,190 records, whilst a sweep for retractions identified 9 records that were excluded from further analysis.

Following screening of titles and abstracts, 19,117 were included in the map, and passed on to annotation. Species annotations were obtained for all 19,117 records, geography for 7,618 records, and topic classifications for 18,886 records.

Flow of records through database searching, deduplication, record repair and enrichment, retraction screening, title-and-abstract screening, annotation and inclusion in the Living Evidence Map.

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Database provenance was retained through deduplication, allowing the contribution and overlap of the five bibliographic sources to be compared before and after relevance screening. Figure 2 shows database contribution to the deduplicated corpus and to the final included set of titles and abstracts, together with the exact source intersections.

Contribution and overlap of bibliographic databases before and after relevance screening. Black bars show all deduplicated canonical records and olive-green bars show records retained after title and abstract screening; the matrix identifies the database combination defining each exact intersection.

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Publication year

The volume of academic evidence on salmon farming increased markedly over time. There were 1,123 included records published in 2025, the highest complete publication year in the current corpus, compared with just 475 in 2010.

The three largest focal species categories were Rainbow trout (7,820 records; 40.9%), Atlantic salmon (7,368; 38.5%) and Unspecified species (3,900; 20.4%). Species categories are not mutually exclusive because individual records can refer to more than one eligible farmed species.

Records by publication year and focal farmed species.

Records by publication year and focal farmed species.

The thematic composition of the literature also expanded through time. The largest high-level topic was Production, followed by Environment. Because records can receive more than one high-level topic, annual stacked totals represent topic-record assignments rather than mutually exclusive studies.

Topic-assignment frequency for high-level topics by
publication year.

Topic-assignment frequency for high-level topics by publication year.

Geography

A substantive study country could be resolved for 7,618 records (39.8%), while 11,499 included records (60.2%) did not contain sufficient evidence for a defensible country-level assignment. Among records with assigned geography (Figure 5), the largest country evidence bases were Norway (1,752 records), Canada (966), Chile (947), United Kingdom (820), United States (561), Turkey (413). Countries demonstrated strong species-specificity, as expected (Figure 6), with the major Atlantic salmon producing countries contributing most to the evidence base, and a large number of other countries contributing collectively to rainbow trout research (e.g. Turkey). Records may be assigned to more than one country where the substantive study spans multiple countries.

Global distribution of included records by substantive study
country.

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Records per country for the 20 most frequently represented
study countries, by focal farmed species.

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Topics

At least one topic was assigned to 18,886 of the 19,117 included records; 231 records (1.2%) remained uncoded for topic. Across coded records, the final ontology contained 53,288 distinct record-topic assignments.

The evidence base was strongly oriented towards Production, represented in 13,984 records (73.1%). Environment was the second-largest theme (4,657; 24.4%), followed by Methods (2,436; 12.7%), Product (2,360; 12.3%) and Industry and governance (2,306; 12.1%). People and society (936; 4.9%) and Inputs and resources (747; 3.9%) were substantially smaller. Theme percentages are not mutually exclusive because a record can be assigned to more than one theme.

High-level topic frequency by focal farmed species. Numbers at bar ends are unique records assigned to each high-level theme; stacked species segments may sum above the unique-record total for multi-species records.

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Production

Production research was centred on fish performance and biology, fish health, feed and nutrition, breeding and reproduction, and stress and mortality. The most frequently assigned terminal topic was Physiology, metabolism and biological responses (4,617 records; 24.2%), followed by Growth and development (2,787; 14.6%). Other major concentrations included Other diseases - Prevention and treatment (1,756; 9.2%), Other diseases - General (1,363; 7.1%), Alternative feed ingredients (1,358; 7.1%) and Other diseases - Epidemiology (1,337; 7.0%). Sea lice remained a substantial research cluster, including 642 records on Sea lice control and treatment and 504 on Sea lice epidemiology.

Topic-assignment frequency within the Production theme.

Topic-assignment frequency within the Production theme.

Environment

Environmental research was concentrated in Physiology, metabolism and biological responses (4,617; 24.2%), Growth and development (2,787; 14.6%), Other diseases - Prevention and treatment (1,756; 9.2%) and Other diseases - General (1,363; 7.1%).

Topic-assignment frequency within the Environment theme.
Topic-assignment frequency within the Environment theme.

Methods

Methods research was dominated by Physiology, metabolism and biological responses (4,617; 24.2%), followed by Growth and development (2,787; 14.6%), Other diseases - Prevention and treatment (1,756; 9.2%) and Other diseases - General (1,363; 7.1%).

Topic-assignment frequency within the Methods theme.
Topic-assignment frequency within the Methods theme.

Industry and governance

Within Industry and governance, Physiology, metabolism and biological responses was the largest terminal topic (4,617; 24.2%), followed by Growth and development (2,787; 14.6%), Other diseases - Prevention and treatment (1,756; 9.2%), Other diseases - General (1,363; 7.1%) and Alternative feed ingredients (1,358; 7.1%).

Topic-assignment frequency within the Industry and governance theme.
Topic-assignment frequency within the Industry and governance theme.

Product

Product research was led by Physiology, metabolism and biological responses (4,617; 24.2%), Growth and development (2,787; 14.6%) and Other diseases - Prevention and treatment (1,756; 9.2%). Other diseases - General and Alternative feed ingredients were also prominent.

Topic-assignment frequency within the Product theme.
Topic-assignment frequency within the Product theme.

People and society

People and society was much smaller than the Production and Environment evidence bases. The largest terminal topics were Physiology, metabolism and biological responses (4,617; 24.2%), Growth and development (2,787; 14.6%) and Other diseases - Prevention and treatment (1,756; 9.2%). Topics relating to Local and rural development, Livelihoods and income, Indigenous peoples and rights, Employment and job creation, Food availability and nutritional security, Culture, identity and traditions, Working conditions and occupational health, Distribution of benefits and harms, Local and Indigenous knowledge, Displacement, disruption and wellbeing, Multiple or general social impacts, Environmental justice and human rights, Competition with other food uses, Health impacts of

aquaculture, Labour rights, exploitation and migration, and Affordability and access each represented under 0.5% of the full evidence map.

Topic-assignment frequency within the People and society theme.

Topic-assignment frequency within the People and society theme.

Inputs and resources

Inputs and resources was the smallest high-level theme. Physiology, metabolism and biological responses accounted for 4,617 records (24.2%) and was more common than Growth and development (2,787; 14.6%), and Other diseases - Prevention and treatment (1,756; 9.2%). Therapeutants and chemicals and Affordability and access were represented in only 9 (<0.1%) and 8 (<0.1%) records, respectively.

Topic-assignment frequency within the Inputs and resources theme.

Topic-assignment frequency within the Inputs and resources theme.

Comparison with secondary research

The supplied manuscript draft compares this evidence map with a recent umbrella review of 37 systematic reviews across salmon aquaculture. Recalculation against the final canonical evidence map confirms the principal imbalance identified in the draft. Sea lice control and treatment occurs in 642 mapped records (3.4%), compared with 8 of 37 systematic reviews (21.6%) in the umbrella-review dataset. By contrast, several of the largest areas in the mapped literature have comparatively little systematic-review coverage, including Physiology, metabolism and biological responses (24.2% of mapped records), Growth and development (14.6%), and major disease, environmental and methodological topics.

The social evidence base is particularly weakly represented in the systematic-review literature described in the draft. In the final canonical map, Antimicrobial resistance and One Health accounts for 24.2% of records, Conflict and social licence for 14.6%, and Public discourse and representation for 9.2%; the supplied umbrella-review dataset contains one review on antimicrobial resistance/One Health and none in most of the other social categories. Overall, the comparison continues to indicate that the systematic-review evidence covers a narrower subset of the topics represented across the mapped literature, with particularly strong representation of sea lice and limited coverage of several major physiological, environmental, methodological, governance and social research areas.

Topic-assignment frequency in the supplied umbrella review of 37 systematic reviews across salmon aquaculture. Data and

figure retained from the supplied manuscript draft.

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